

Def. Let $p: E \rightarrow B$ be a continuous onto map.

An open set $U \subset B$ is said to be evenly covered by p if:

$$p^{-1}[U] = \bigcup_{\alpha \in I} V_{\alpha} \text{ where } V_{\alpha} \text{ is disjoint open sets such that}$$
$$\text{for all } \alpha \in I, \quad U \stackrel{p}{\cong} V_{\alpha}$$

The map $p: E \rightarrow B$ is called a covering map if:

for each $b \in B$, there exists an evenly covered neighbor U containing b .

If such a map p exists, then E is called the covering space.

Prop. Every covering map is open map.

Proof. Let $p: E \rightarrow B$ be a covering map, and $W \subset E$ be an open set.

To show $p[W]$ is an open set, let $b \in p[W]$ be given.

Since the p is covering map, there exists an evenly covered open set U containing b . By definition, for some disjoint open sets V_{α} ($\alpha \in I$),

$$p^{-1}[U] = \bigcup_{\alpha \in I} V_{\alpha} \quad b \in U$$

and, $b \in p[W]$, so $b = p(y)$ for some $y \in W$. Since $b \in U$, so $p^{-1}[p[\{y\}]] \subset p^{-1}[U]$, therefore $y \in p^{-1}[U] = \bigcup_{\alpha \in I} V_{\alpha}$, thus $y \in V_{\alpha_0}$ for some $\alpha_0 \in I$.

Now, $V_{\alpha_0} \cap W$ is an open set of E containing y , so that $V_{\alpha_0} \cap W$ is an open set of V_{α_0} .

Since the restriction

$$p|_{V_{\alpha_0}}: V_{\alpha_0} \rightarrow U$$

is Homeomorphism, so $p[V_{\alpha_0} \cap W] \subset U$ is an open set of U , and since U is open set of B , thus $p[V_{\alpha_0} \cap W]$ is open set of B .

Finally, $b = p(y) \in p[V_{\alpha_0} \cap W] \subset p[W]$, so $p[W]$ is an open set of B . $\square$

Lift.

Def. Let $p: E \rightarrow B$ be a Continuous map.

For a Continuous map $f: X \rightarrow B$, a lifting of f is:

a Continuous map $\tilde{f}: X \rightarrow E$ such that $p \circ \tilde{f} = f$.



Lemma. Let $p: E \rightarrow B$ be a Covering map and $f: X \rightarrow B$ be Continuous maps.

If there exist two lifts $\tilde{f}_1, \tilde{f}_2: X \rightarrow E$ of f with $\tilde{f}_1(x_0) = \tilde{f}_2(x_0)$ for some $x_0 \in X$, and X is Connected, then $\tilde{f}_1 = \tilde{f}_2$.

Thm. Let $p: (E, e_0) \rightarrow (B, b_0)$ be a covering map.

For any Path $\alpha: I \rightarrow B$ such that $\alpha(0) = b_0$, there exists a Unique Path

$\tilde{\alpha}: I \rightarrow E$ such that $p \circ \tilde{\alpha} = \alpha$ and $\tilde{\alpha}(0) = e_0$.

Diagrammically,

